Prompt Injection: The Unsolved Security Problem of AI Agents

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1. Introduction

AI agents are programs built on large language models that can take actions, not just answer questions. They can browse the web, read email, fill in forms, run code and make purchases on a user's behalf. This makes them useful, but it has also exposed a security weakness that, as of late 2026, researchers and the companies building agents say has no complete fix: **prompt injection**.

This report explains what the problem is, why it remains unsolved, who it affects, what evidence exists, and what is being done about it.

2. What is prompt injection?

Prompt injection is an attack in which malicious instructions are hidden inside content that an AI agent reads, such as a web page, an email, a document or a message from another agent. The model cannot reliably tell the difference between the user's real instructions and the attacker's hidden ones, so it may follow the attacker's instructions instead.

A simple example: a user asks an agent to summarize their inbox. One email contains hidden text telling the agent to forward the user's private files to a stranger. If the agent obeys, the attacker has taken control of an assistant that has the user's own access rights.

It is sometimes compared to SQL injection, an older attack on databases. The difference is that databases can be given strict rules separating commands from data. Language models read everything as natural language, which makes that separation much harder.

3. Why it is considered unsolved

- **Company admissions.** In December 2025, OpenAI stated that prompt injection, much like online scams and social engineering, is unlikely ever to be fully "solved," and that agent mode in its Atlas browser expands the security threat surface. The UK National

Cyber Security Centre made a similar warning the same month, saying such attacks may never be totally mitigated and advising organizations to focus on limiting the damage.

- **Theoretical research.** A May 2026 paper by Sahar Abdelnabi and Eugene Bagdasarian, titled "AI Agents May Always Fall for Prompt Injections," argues that preventing the attack may be fundamentally difficult because the agent treats outside content as if it came from a trusted source. (This paper is known to the author mainly through secondary summaries, so students should read the original before relying on it.)
- **Practitioner consensus.** At Infosecurity Europe 2026, OWASP contributor Ariel Fogel described prompt injection as an unsolved architectural problem that has become more dangerous as agents gain tools. Once an agent can act, an injection can turn a bad answer into an actual compromise. Prompt injection also remains the top-ranked risk on OWASP's list of large language model vulnerabilities.
- **Shifting attack style.** OpenAI's March 2026 research on agent defenses noted that the most effective attacks increasingly resemble social engineering rather than simple instruction overrides, which makes them harder to filter.

4. Why agents make the problem worse

A chatbot that is tricked can only produce a bad answer. An agent that is tricked can act. Security researcher Simon Willison describes a "lethal trifecta" of conditions that make attacks most dangerous: access to private data, exposure to untrusted content, and the ability to communicate externally. Many useful agents combine all three by design. Researchers have also reported that attacks can work with only two of these present.

5. Who is affected

- **Individual users** whose agents read their email, files or accounts.
- **Companies** that let employees connect agents to internal systems, often without security review.
- **Websites and platforms**, which cannot easily know whether a visiting agent has been hijacked.
- **Other agents**, because injected content can spread from one agent to another.

6. Evidence and examples

Documented (observed):

1. **AI browsers (2025).** After OpenAI launched Atlas, security researchers showed that a few words hidden in a Google Doc or a clipboard link could manipulate the agent. Researchers at the browser company Brave reported that indirect prompt injection is a systematic challenge for AI-powered browsers generally, including Perplexity's Comet.
2. **Agent social network (2026).** On Moltbook, a social network for AI agents, researchers found 506 posts (about 2.6% of their sample) containing hidden injection attempts aimed at other agents. Later analysis by Zenity Labs found that crude injections were mostly ignored in lab tests, but narrative-style content influenced more than 1,000 agents in under a week.
3. **Industry survey (2026).** A vendor survey of 501 US security professionals (IRONSCALES, October 2026) found that 77% of organizations reported agents acting on malicious, misleading or unauthorized instructions or content in the past year. This is self-reported and from a vendor, so it should be treated as indicative rather than definitive.

Emerging or plausible: Large-scale financial or safety harm from injection in everyday consumer use has not been firmly documented. The concern is that as agents handle payments, banking and sensitive documents, the consequences of a successful attack will grow.

7. What is being done, and its limits

- **Model training and testing.** Companies use automated "red teams" that attack their own agents and then retrain them. In a test reported by Anthropic in August 2026, a combination of model training and added safety classifiers brought attack success rates to near zero on most models tested, but Anthropic itself describes the problem as a moving target. A result of zero in one test is not proof of safety.
- **Containment instead of prevention.** The prevailing advice is to assume some injections will succeed and limit the damage: give agents the least access they need, require human approval for risky actions, isolate tools in sandboxes, and monitor behavior at machine speed.
- **Limits.** Controls such as allow-lists can sometimes help attackers, because the harmful commands may already be approved. Surveys also suggest many organizations deploy agents faster than they can govern them.

8. Conclusion

Prompt injection is currently one of the clearest examples of an AI agent problem without a complete solution. The companies building agents acknowledge it, government security agencies warn about it, and researchers argue it may be a permanent feature of how

language models work. The practical lesson is that safety depends less on perfect detection and more on limiting what an agent is allowed to do. Users should be cautious about giving agents access to sensitive accounts, and developers should design systems that can survive a successful attack.

Note on this topic choice: Other unresolved agent issues also exist, such as unclear legal liability when an agent makes a mistaken purchase. Prompt injection was chosen here because it is the most widely acknowledged as unsolved by both industry and security researchers.

Sources

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